Replicating Braboszcz et al. (2017) on OpenNeuro `ds003969`: pipeline decisions, sensitivity, and a look at criticality

October 2026

Abstract

I rebuilt the EEG pipeline of Braboszcz et al. (2017, *PLoS ONE* 12:e0170647) in MNE-Python and applied it to the public release of their data. The alpha result reproduces: Vipassana practitioners have more 7–11 Hz power than the other meditation groups during meditation, and this holds under every pipeline variant tried. The headline gamma (60–110 Hz) excess in meditators over controls is less robust. Its sign always agrees with the paper, but it is clearly significant only when no ICA components are removed. Automated muscle-component removal roughly halves it. I then computed long-range temporal correlations (DFA) and functional E/I (fEI) with the parameters of the recent preprint "Near-critical brain dynamics track effortlessness during meditation". As in that preprint, there are no meditator–control differences. The one apparent within-meditator effect (beta DFA higher in the second meditation block than in the first) also appears, and is larger, in controls. It is therefore best read as a time-on-task effect, not as an effect of meditative depth.

1 Data and target results

The dataset (`ds003969`) contains 64-channel Biosemi recordings of 98 participants: 31 controls (CTR), 24 Himalayan Yoga (HTR, "HYT" in the paper), 20 Isha Shoonya (SNY, "ISY"), 19 Vipassana (VIP) and 4 Transcendental Meditation (TM). Each participant did a 20-minute instructed mind-wandering (IMW) session and a 20-minute meditation session (MED). These are released as four ∼10-minute files: `think1`, `think2`, `med1breath` (breath focus) and `med2` (the tradition's own practice; controls keep doing breath focus). The paper analysed the last 10 minutes of each session, which here are `think2` and `med2`.

The results targeted were:

1. **Gamma (60–110 Hz).** Higher parieto-occipital power in all three meditator groups than in controls, both as a trait (MED and IMW combined) and during MED (Fig. 3b).
2. **Alpha (7–11 Hz).** During MED, VIP higher than CTR, ISY and HYT.
3. (Not attempted) the correlation between gamma and log practice hours (Fig. 4D).

2 Pipeline: decisions and likely discrepancies

Table 1 sets the paper's description against what was implemented. Every step was either matched as closely as the text allows or replaced by an automated equivalent, because the original used manual judgement.

Where results are most likely to diverge. In rough order of expected impact:

1. **ICA component selection.** The paper removed 1–5 components chosen by eye. The first automated run removed more than 10 of 20 components in many recordings (Fig. 4a). Section 3 shows that this step controls the gamma result.
2. **Sample composition.** The replication uses about 50% more participants and does not age-match. HTR is on average 11 years older than CTR. High-frequency EEG power and EMG both change with age, so the gamma contrast for HTR is confounded with age in a way the paper's matched design avoided.
3. **Bad channels.** The automated detector removes slightly more channels than the paper did (6.9 vs ≈5 on average). Channels are dropped, not interpolated, so the parieto-occipital average is taken over a cluster whose make-up varies from subject to subject.
4. **Statistics.** The cluster permutation approach was not re-implemented. The tests here are uncorrected tests on one predefined cluster, so their p-values are not directly comparable to the paper's.
5. **Unreproducible manual steps.** The paper's `pop_rejcont` threshold tweaks and its channel and component choices cannot be recovered from the text.

Step	Paper	This replication
Sample	16 per group, matched on age and gender from a larger pool; TM not included	All 94 non-TM participants, unmatched (mean age HTR 50.7 vs CTR 39.8; SNY 18/20 male)
Reference	Right mastoid	Right mastoid (`EXG6`). `Fp1` replaced by `EXG8`, the spare electrode wired in its place
Resample, filter	256 Hz; 1 Hz IIR high-pass, order 6, 0.3 Hz transition	Same
Segment rejection	EEGLAB `pop_rejcont`: 1 s epochs, 0.5 s overlap, 0–10 Hz >17 dB, 35–128 Hz >14 dB, contiguous 8; thresholds "in rare cases" adjusted by 1–5 dB after visual review	Line-by-line Python port with unit tests, including EEGLAB quirks such as `contiguous` acting as a merge distance. Manual threshold adjustment **not reproduced**
Bad channels	Manual; 0–18 per subject, mean ≈5; interpolation not stated	Automated: union of MNE waveform LOF (thr. 1.5) and a custom spectral LOF on 1–45 Hz log-PSD (thr. 2). Mean 6.9, median 6, range 0–31. Not interpolated; bad channels are excluded and cluster means use the remaining channels
ICA	Infomax; 1–5 eye/temporal-muscle components chosen visually by one experimenter	Infomax with 20 components (count not given in the paper), fit on good EEG channels. Components chosen automatically with `find_bads_eog` and `find_bads_muscle`; thresholds and caps varied across runs (Table 2)
Spectra	1 s non-overlapping epochs, Hanning taper, FFT, $10\log_{10}$; 2–110 Hz, 50 Hz band excluded	Same epoching and taper (Welch with one segment per epoch); epochs overlapping `BAD` spans dropped; linear power averaged over epochs, then dB
Statistics	FieldTrip cluster permutation on log power (2000 resamples) over electrodes; Welch/mixed ANOVA	Welch t-tests on the mean over an 8-electrode parieto-occipital cluster (O1, Oz, O2, PO7, PO3, POz, PO4, PO8); no cluster correction

Table 1: Paper pipeline compared with the replication.

Two smaller points came up along the way:

- The paper's y-axis (∼45 dB at low frequencies) is offset from a correctly scaled PSD in $\mu V^2/Hz$ by a constant of roughly 35–45 dB. This is consistent with an unnormalised squared FFT magnitude. Because it is a constant offset in dB, between-group differences are unaffected.
- The 100 Hz line-noise harmonic falls inside the gamma band and is not excluded. Visual inspection showed nothing abnormal at 100 Hz.

3 Sensitivity of the headline results

Gamma. Across every variant the meditator-minus-control difference is positive for all three groups, so the direction of the paper's result reproduces. Its size and significance depend on ICA:

- **No ICA:** trait differences of +3.3 dB (HTR, $p = .002$), +2.2 dB (SNY, $p = .03$) and +1.9 dB (VIP, $p = .12$). During MED, HTR and SNY are both significant at $p \leq .02$. This is close to the paper's picture.
- **Automated ICA:** the differences shrink to 0.6–2.0 dB and are at most marginal. VIP is hit hardest, falling from +2.2 to ≈0 dB during MED.

The log's working hypothesis (8 Oct) was that EOG component removal drives the discrepancy. The data point the other way. Raising the EOG threshold removes about 0.9 fewer EOG components per recording but changes gamma little: HTR trait goes from 1.6 to 2.0 dB and VIP from 0.8 to 0.6 dB. Most of the gap between the ICA runs and no ICA is explained by the ∼2.7 *muscle* components per recording that remain. The number of muscle components removed is similar across groups (2.0–2.6 per recording), so the attenuation does not come from removing more components in meditators. It comes from *what* those components contain.

Two readings fit this:

1. The "muscle" detector correctly removes EMG that is more prevalent in meditators. The paper's gamma effect would then be partly artefactual.
2. The detector also captures genuine broadband high-frequency activity, or microsaccadic spike potentials, which a spectral-slope muscle criterion cannot tell apart from EMG.

The paper's own control analysis is relevant here: its eye-component cluster correlated with parieto-occipital

Run	ICA rule	ICs removed per recording	Gamma at Oz, `med2`
v1†	muscle thr. 0.5, EOG z>3, uncapped	often >10 of 20	No CTR deficit (Fig. 3a)
v2†	muscle thr. 1.0, EOG z>5	0 in practice	CTR clearly lowest (Fig. 3c)
v3†	muscle 0.75, EOG z>3, uncapped	median ∼2; many 0; tail to 17	No CTR deficit
capped	muscle 0.75, EOG z>3; ≥1 muscle, ≤4 EOG, ≤5 total	3.4 (2.4 muscle, 1.0 EOG)	Right direction, not significant (Fig. 3d)
capped + EOG prefilter	as above, with EOG channels filtered beforehand	3.3	Same as capped
capped, EOG z>4	as above but EOG z>4	2.8 (2.7 muscle, 0.1 EOG)	See Fig. 1
No ICA	none	0	See Fig. 1

Table 2: Pipeline variants. †Early runs whose outputs were later overwritten; results are from the project log. The capped run is in `outputs/`; the capped + EOG prefilter run is in `outputs_v2/`; the capped, EOG z>4 run is in `outputs_v3/`, and `outputs_v4/` is byte-identical to it; the no-ICA run is in `outputs_noica/`. The log describes the `_v3` rerun as "exclude bad channels from ICA". That change is a no-op, because `ICA.fit` already excludes bad channels by default (checked by refitting one recording). The drop in EOG components comes from the EOG threshold rising from 3 to 4, an uncommitted change in the working tree. Because `_v3` already used z>4, the planned "`v4`: higher EOG threshold" run duplicated it.

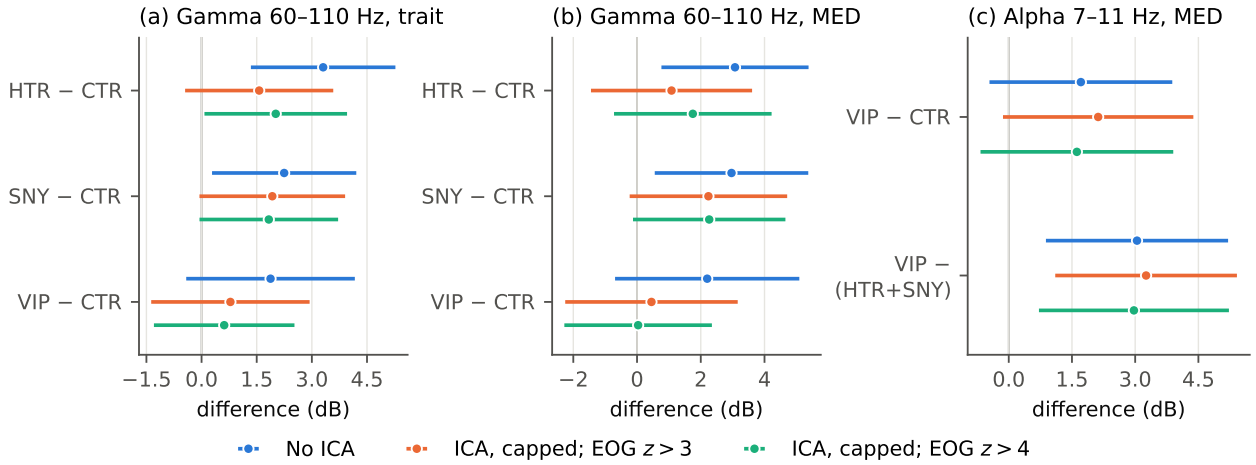


Figure 1: Group differences in parieto-occipital cluster power (dB) under three pipelines. Points are differences in group means and bars are 95% CIs from Welch t-tests. Trait = each subject's mean over `think2` and `med2`; MED = `med2`. The capped + EOG prefilter run is omitted because it lies within 0.1 dB of the capped run (EOG z>3).

gamma ($r = 0.33$), whereas temporal-muscle components did not. The automated pipeline cannot settle this question. Careful manual labelling of a subset of recordings could; this was started on 8 Oct (19 recordings in `processed_manual`). Early impressions were that the muscle suggestions look sensible and the EOG suggestions often need correcting.

Alpha. VIP minus the other meditators during MED is +3.0 to +3.3 dB (p = .006–.014) under every pipeline, and is similar for IMW and for the trait average. This part of the paper's alpha result is robust. VIP minus CTR is +1.6 to +2.1 dB but not significant (p = .07–.17), whereas the paper reports $p < .01$. The likeliest explanations are the unmatched and larger control group, and the cluster-level statistics.

4 Criticality: DFA and fEI

Method. The Hilbert amplitude envelope was computed in delta, theta, alpha (8–13 Hz), beta (13–30 Hz), broadband (1–30 Hz) and (no-ICA run only) gamma. DFA and fEI used `crosci` on non-overlapping 180 s epochs, with DFA fitted over 5–30 s and fEI computed in 5 s windows with 80% overlap. Values were averaged over epochs and good channels to give one value per subject, task and band. Manual implementations of DFA and fEI were first written and checked against `crosci`; they differed slightly because of windowing.

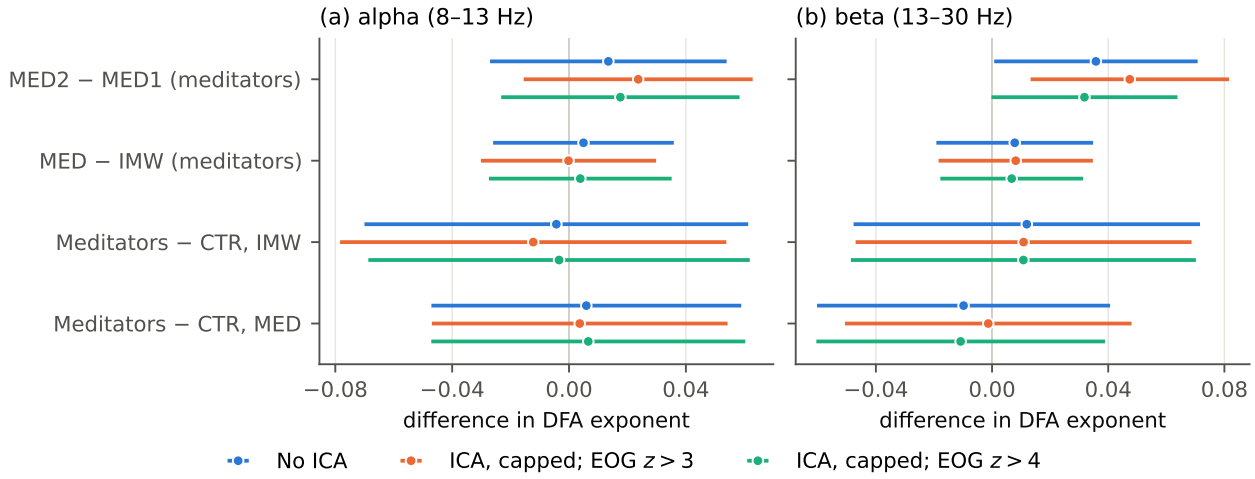


Figure 2: DFA contrasts (mixed-model estimates, 95% CIs) under three pipelines. "Meditators" means HTR, SNY and VIP pooled. MED − IMW pools both blocks of each condition.

Contrasts were fitted as linear mixed models with a random intercept per subject. The project notebook used WLS with HC3 errors and treated subject–task rows as independent; the conclusions are the same. TM was excluded.

Data loss. An epoch is dropped if *any* `BAD` span overlaps it. Even a 2 s `pop_rejcont` flag therefore discards 180 s of data. Consequences:

- Only 232 of 392 recordings have at least one usable epoch (222 non-TM, from 83 subjects).
- `think2` retains only 43 of 98 recordings.
- Usable recordings average 1.7 epochs, about 5 of 10 minutes.
- Only 37 meditators have both meditation blocks.

This is the most important practical weakness of the criticality analysis. Possible fixes are 50%-overlapping windows (as in the preprint), computing DFA on concatenated clean data, or a less aggressive rejection rule. An epoch-length sweep (40–300 s, `exploration/261012_*`) has been written, but the saved output covers only 2 recordings and is not yet informative.

Results (Fig. 2).

- **MED vs IMW in meditators:** no difference in any band (alpha and beta estimates < 0.01, $p > .5$).
- **Meditators vs controls:** no difference during IMW or MED in any band from theta to broadband ($|\Delta| < 0.02$, $p > .5$). The CIs rule out differences larger than about 0.06–0.08 DFA units, roughly 10% of the mean exponent of ≈0.7.
- **fEI:** nothing reaches $p < .05$ in any contrast or band except gamma.
- **`med2` vs `med1breath` in meditators:** beta DFA is higher in the second block, by +0.032 to +0.048 ($p = .007$–$.051$ depending on pipeline). Alpha shows the same direction but is not significant.
- **Gamma (no-ICA run only):** DFA and fEI are higher in meditators than controls during MED ($p = .013$, $p = .028$). Given the ICA sensitivity of gamma power above, this should be treated as artefact-prone until shown otherwise.

The `med2` effect is probably time-on-task. `med2` always follows `med1breath`, and controls do breath focus in both blocks. If the rise reflected deepening practice, controls should not show it. They do, and more strongly:

- Beta DFA rises by +0.075 to +0.077 in controls ($p = .005$–$.009$, $n = 28$).
- Alpha DFA rises by +0.085 to +0.094 in controls ($p = .02$–$.05$).
- The block × group interaction is not significant for beta, and for alpha it is *negative* (meditators rise less).

DFA also rises from `think1` to `think2`: for meditators' alpha by about +0.05 ($p = .02$–$.03$), and in a smaller set of controls by a similar amount. The simplest account is that LRTCs in the alpha and beta envelopes increase over the course of any 20-minute eyes-closed block. Drowsiness is the obvious candidate mechanism.

Multiplicity reinforces this reading. About 40 contrasts were examined per pipeline (5 bands × 2 metrics × 4 contrasts) across 4 pipelines. A single $p \approx .007$ would not survive a Bonferroni or BH correction.

5 Comparison with "Near-critical brain dynamics track effortlessness during meditation"

Lewis-Healey, Kringelbach, Canales-Johnson and Laukkonen (bioRxiv, Aug 2026) recorded 128-channel EEG from 33 TM practitioners and 33 matched controls. Sessions consisted of a 10-minute rest, then 30 minutes of TM or silent counting, then a further 10-minute rest. Effort was rated retrospectively over time with Temporal Experience Tracing. Their criticality settings are essentially those used here: `crosci`, the same five bands, 3-minute windows, DFA up to 30 s, fEI with 5 s windows at 80% overlap, and fEI only where DFA > 0.6. They differ in a few respects:

- 50% window overlap.
- Band-specific minimum DFA windows (alpha 3.98 s, beta 2.24 s).
- Average reference, 0.5–45 Hz FIR filtering, FastICA with manual component rejection.
- Rejection on 5 s epochs.

The dataset is different. The TM group in `ds003969` has only 4 participants and was not analysed here.

Agreement. The preprint finds *no robust between-group differences* in LRTCs or fEI, during either rest or practice. The replication agrees across all bands and pipelines, with three meditation traditions instead of one. The preprint also finds no fEI condition effects after FDR correction, which matches the null fEI results here.

Apparent disagreement. Within meditators, the preprint finds TM increases alpha and beta DFA relative to rest both *before and after*. Here, MED does not differ from IMW. The comparisons are not equivalent:

- IMW is an active, effortful recall task rather than rest, so it is a weaker contrast for a hypothesis about "letting go".
- The preprint's pre/post bracketing controls for slow drift. The MED-vs-IMW comparison here does not; session order is counterbalanced only at the group level, and much data has been lost.

The bracketing design matters. The `med2`-vs-`med1breath` analysis above shows that drift within a block is large enough (about +0.08 DFA units over 10 minutes in controls) to produce a spurious "deepening" effect when blocks are not counterbalanced.

What this data can and cannot say about effortlessness. The preprint's main finding is within-person: beta LRTCs rise as reported effort falls, more so in meditators. The strongest effect is beta (standardised $\beta = -0.28$), and meditators show higher beta LRTCs only in their lowest-effort windows. `ds003969` has no time-resolved phenomenology, so this cannot be tested directly. The nearest proxy, the move from effortful breath focus to the tradition's own practice, does show a beta-DFA increase. It is not specific to meditators, however, so it cannot be attributed to effortlessness. Two points follow:

- Time-on-task drift is a plausible confound for any effort–LRTC association, because reported effort also tends to fall over a session. The preprint addresses this with time-control models, and its effects survive (alpha $\beta = -0.47$, beta $\beta = -0.50$).
- The size of the drift seen here ($\sim$0.08 in 10 minutes) is a useful benchmark for how strong that control must be.

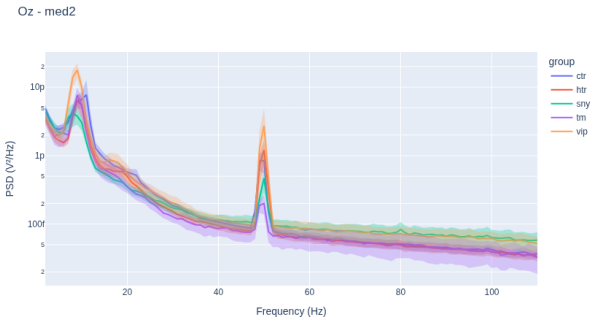
6 Summary and next steps

- **Alpha (VIP > other traditions): replicates**, and is insensitive to every pipeline decision tried.
- **Gamma (meditators > controls): direction replicates; strength does not.** It is significant without ICA, but automated muscle-component removal halves it. The cause is the content of the removed components, not how many are removed.
- **Criticality: null between groups**, consistent with the near-criticality preprint. The one within-meditator effect is most parsimoniously explained by time-on-task.

Next steps, in priority order:

1. Re-run the spectral comparison on an age- and gender-matched subsample of 16 per group.
2. Manually label ICA components for a stratified subset of recordings and compare the gamma result with the automated choices, looking specifically for saccadic spike-potential components.
3. Reduce data loss in DFA/fEI with overlapping windows or concatenated clean segments, and finish the epoch-length sweep.
4. Replace the uncorrected tests with cluster permutation tests, or at least FDR correction across bands and contrasts.

A Figures from the project log



(a) First run (aggressive ICA).

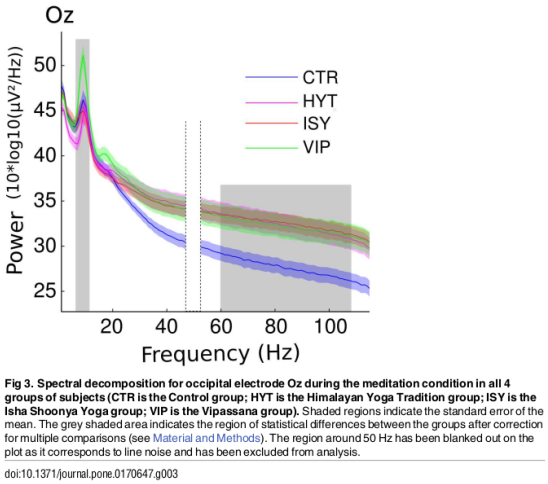
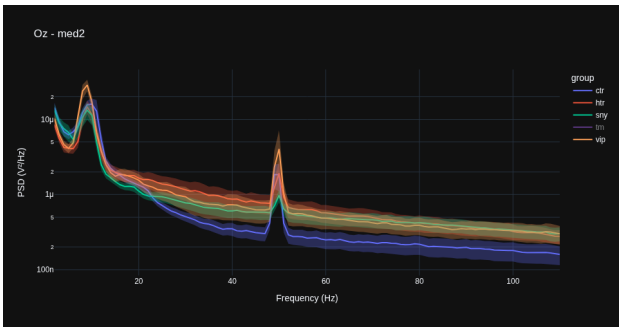


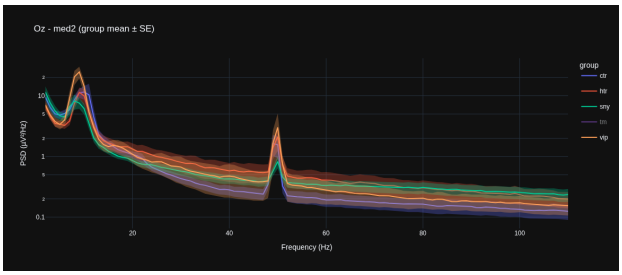
Fig 3. Spectral decomposition for occipital electrode Oz during the meditation condition in all 4 groups of subjects (CTR is the Control group; HYT is the Himalayan Yoga Tradition group; ISY is the Isha Shoonya Yoga group; VIP is the Vipassana group). Shaded regions indicate the standard error of the mean. The grey shaded area indicates the region of statistical differences between the groups after correction for multiple comparisons (see Material and Methods). The region around 50 Hz has been blanked out on the plot as it corresponds to line noise and has been excluded from analysis.

doi:10.1371/journal.pone.0170647.g003

(b) Paper, Fig. 3.

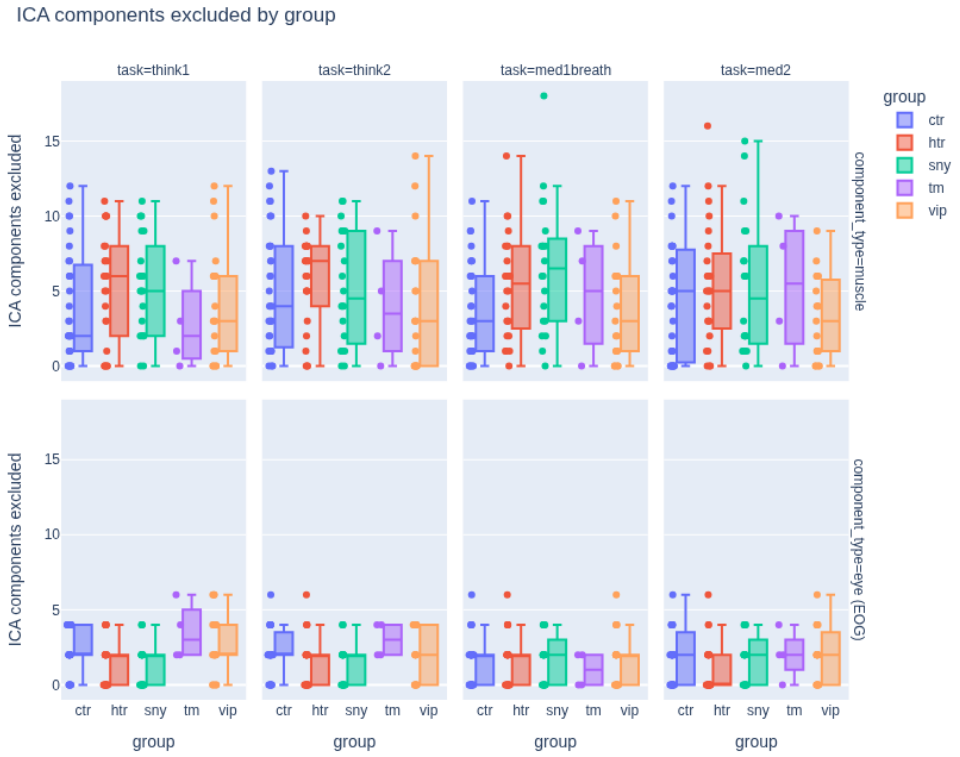


(c) Run with effectively no ICA removal.

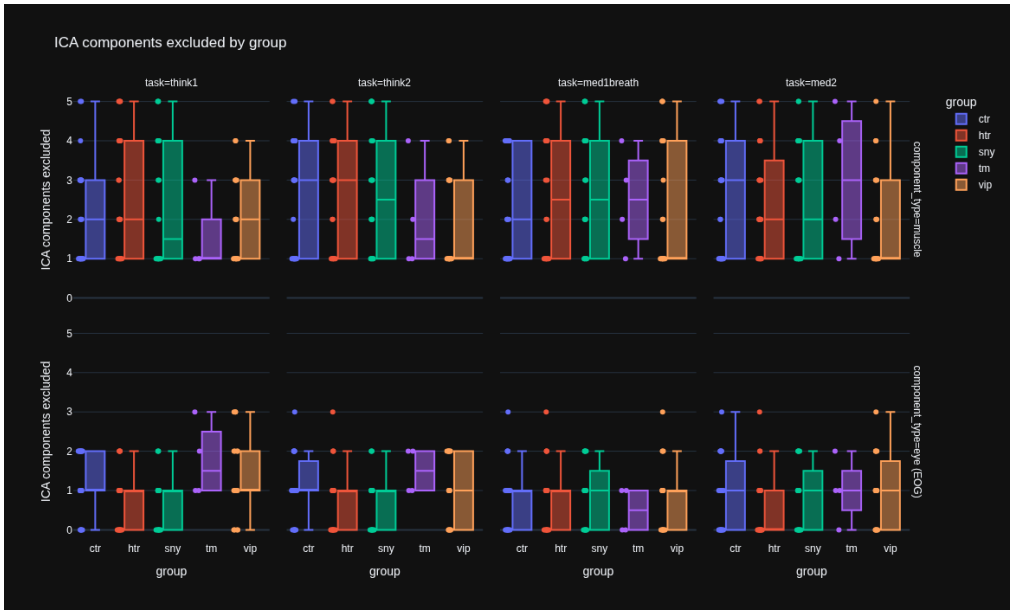


(d) Capped ICA run.

Figure 3: Oz spectra during `med2`, group mean ± SE, under successive pipelines, alongside the paper's figure.



(a) First run.



(b) Capped run.

Figure 4: ICA components removed per recording, by group and task, split into muscle (top) and EOG (bottom).

References

- Braboszcz C, Cahn BR, Levy J, Fernandez M, Delorme A (2017). Increased gamma brainwave amplitude compared to control in three different meditation traditions. *PLoS ONE* 12(1): e0170647. `https://pmc.ncbi.nlm.nih.gov/articles/PMC5261734/`
- Lewis-Healey E, Kringelbach M, Canales-Johnson A, Laukkonen R (2026). Near-critical brain dynamics track effortlessness during meditation. *bioRxiv*. `https://doi.org/10.64898/2026.08.07.743470`
- Dataset: OpenNeuro `ds003969` v1.0.0. `https://openneuro.org/datasets/ds003969/versions/1.0.0`